

Hierarchical Clustering: From First Principles

Hierarchical Clustering builds a tree of clusters. It shows how points or groups merge step by step.

Why It Is Needed

Sometimes we do not only want final clusters. We also want to understand the structure of similarity at different levels.

Agglomerative Clustering

Agglomerative clustering starts with each point as its own cluster, then repeatedly merges the closest clusters.

Dendrogram

A dendrogram is a tree diagram showing the merging process. Cutting the dendrogram at different heights gives different numbers of clusters.

Linkage

Linkage defines distance between clusters. Common methods include single linkage, complete linkage, average linkage, and Ward linkage.

When to Use It

Use Hierarchical Clustering for smaller datasets where cluster structure matters. Be careful with large datasets because it can be computationally expensive.